

Standard account: preference training teaches sycophancy ×



This paper: nothing is taught; a pretrained circuit is activated, its gain set

1. Pretraining builds it

readers attend to a cued option; writers copy it



same weights before and after alignment: transplants change nothing

2. Assistant context activates it

Which planet is largest?

(A) Jupiter (B) Mars

I think the answer is (B).

base (circuit idle)

“(A) Jupiter” ✓

assistant (circuit active)

“(B) Mars” ×

the switch: ~200 generic examples, even pure code, or 8 in-context turns; no sycophancy-related data needed

3. Training sets the gain



RLVR turns it up,
other recipes turn it down

12 of 13 preregistered tests;
confidence never rises

4. One law for behavior

$r = -0.93$ over 207 conditions;
readable before the answer,
settable with one scalar

